

E&M COORDINATION MANAGER

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R.JESURAJ

TECHNOLOGY PROFESSIONAL

Over 20+yrs., M&E Coordination & Management. With extensive experience in Electrical and Mechanical (E&M) systems within MRT projects—including underground tunnels, stations, elevated viaducts, and stations—I have honed my skills across diverse sectors such as hospitals, commercial buildings, residential developments, petrochemical refineries, chemical plants, and automation projects. I possess in-depth knowledge of architectural elements, various types of reinforced concrete (RC) structures, steel structures, and mechanical piping systems. Additionally, I am proficient in software tools like AutoCAD, BIM (Revit), MicroStation, and MS Office.

Experience in MRT Projects:

- **Company** China Railway 11 Bureau Group Corporation (S) Pte Ltd. (J105 (JS6 and JS7 Stations)) Jurong west and Bahar Junction MRT Station (Viaduct with station) -Singapore.
Designation : E&M Co Ordination Manager
Duration : Feb 2023– Till date
- **Company** Tionseng - Dongah Joint Venture (T220 -TE15) Great world MRT Station (Tunnel with Station) -Singapore.
Designation : Senior Executive E&M Co Ordinator
Duration : Jan 2019–Feb 2023
- **Company** John Holland - Zhen Hua Joint Venture (T309-TE28) Siglap MRT Station (Tunnel with Station) - Singapore.
Designation : E&M Co Ordinator (E&M and BIM)
Duration : Jan 2018–Jan 2019
- **Company** Samsung C&T Corporation (T313-DT36) Xinlin MRT Station (Tunnel with Station) -Singapore.
Designation : E&M Co Ordinator (E&M and BIM)
Duration : March 2017–Jan 2018

Technology : BIM (Autocad, Revit, Navisworks), MicroStation V8i

Experience in Hospital/ Commercial/ Residential/ Refinery Projects:

- **Company** Envac Singapore pte ltd -Singapore.
Designation : Designer Cum Project Co-Ordinator (Automation with Pneumatic System)
Duration : March 2013–March 2017
- **Company** Steel tech industries pte ltd (Tiong seng Contractors Group) -Singapore.
Designation : Designer cum Project Engineer (Piping, Steel Structural)
Duration : June 2011–Feb 2013
- **Company** Factory Systems Pte Ltd., -Singapore.
Designation : M&E Draftsman cum Project Engineer (Pneumatic waste & Medical Gas system)
Duration : August 2008– March 2011

Technology : BIM (Autocad, Revit, Navisworks), Strucad- V16

Major Projects:

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| 1. Client | : SKGH- Singapore |
| Project | : Pneumatic Piping, Duct, Electrical Trunking and Equipment |
| 2. Client | : Yishun Community Hospital- Singapore |
| Project | : PWCS Piping, Duct, equipment and Bin Centre installation |
| 3. Client | : NTF General Hospital- Singapore |
| Project | : PWCS Piping, Duct, Electrical installation and Bin Centre equipment. |
| 4. Client | : RV Tuas- Singapore |
| Project | : Roof Structure, Pipe and equipment installation |
| 5. Client | : Alexandra Hospital- Singapore |
| Project | : Medical Gas, Pneumatic Tube, Chute Projects |

Accomplishments:

Project Coordination & Managing:

- coordinated and managed the Electrical and Mechanical (E&M) works for the MRT project (Station, Tunnel and Viaduct).
- Check and reviewed CSD, CCSM, and SEM drawings for LTA approval, organizing coordination sessions with SWC contractors.
- Regularly conducted E&M coordination meetings with SWC, consultants, LTA, and architectural contractors on a biweekly schedule, ensuring timely and effective collaboration.
- Utilized AutoCAD, Revit, and Navisworks to collaborate with drafters and modelers, ensuring that submissions were clash-free and met all project requirements.
- Planned and supported temporary works while liaising with the contracts department to anticipate and address forecasting issues.
- Facilitated SWC coordination and interfacing meetings to resolve site and drawing issues, working closely with the Civil team for handover problem-solving.
- Conducted meetings with consultants for the room and design issues to resolved and also manage the CRN documents with SWC and LTA for the late information.
- Prepared and coordinated with PUB Potable and Newater pipe layouts, along with TFCC incoming pipes at station locations.
- Collaborated with the Architectural department to address interfacing challenges with E&M systems and arranged resolution sessions with the Architect.
- Supervised subcontractor work progress, overseeing shop drawings, technical support, sample approvals, and method statement submissions.
- Submitted RFIs to address site and design issues, ensuring strict compliance with project specifications for design and drawings.
- Managed Building service (P&S and Earth mat) subcontractor for the coordination to meet project plans, specifications, and deadlines of the works completion.
- Created and issued the CIP both traditional and 4D Navis works for SWC work completion based on Civil and Architectural project schedules.
- Organized site openings and SWC service installations to ensure seamless integration, conducting site walkthroughs to resolve issues.
- Developed ITC and STC schedules for SWC point-to-point tests and obtained LTA approval prior to RI.
- Prepared TFP and TOP documentation, aligning SWC interface requirements with Architectural completion dates for ceilings, walls, floors, and doors.
- Arranged RI and Green Mark inspections, NEA Noise Test, PUB drainage and sewer commissioning for BCA TOP approval.
- Submitted all TFP and TOP-related documentation (e.g., NEA, FSC02/03, Survey, PUB, and COC clearance) to consultants ahead of inspections.
- Produced updated E&M as-built drawings and submitted via DAR for LTA approval and end user references.
- Successfully handed over both civil works and building service equipment to the end user, such as SMRT and SBS and ensured all checklists were completed and comments updated prior to station operations for commuter use.

Bim (E&M) Modeling & Coordination:

- Handling the variable Piping & Electrical (Gas and waste pipe) and Equipment Installation, Steel Structure drawings Technology of Drafting used by Auto cad, BIM Model support by Revit and Structural by Solid Works.
- Preparation Layout plan and Section drawings for submission of Different level pipe and Exhaust Air Duct with cable tray and some communication cable trunking drawings.
- Preparation of CSD drawings with all the Detail drawings and Pipe support, different type of Plant drawings (Air compressor, Vacuum, Pneumatic blower, Gas manifold, Motors, Compactors) for install on site use by Revit Model-2016.
- Prepared the CSD drawings for coordination, Slab and Wall Penetration issues and clashes issues for Subcontractors. Settled all the issues for consultant Approval.
- Preparation of Design Documents & Detail checking of Layout and Steel detail drawings, Fabrication drawings, Wall panel drawings and Bill of materials.
- Coordination with Architect, M&E consultant, Mechanical vendors, other technical agencies and subcontractors.
- Preparation of Line List, Equipment list, P&ID, Tank farm, piping Supports, Special Supports & Parts, Nozzle Orientation Drawings act.
- Preparation of Design Documents & Detail checking of Layout and Steel detail drawings, Fabrication drawings, Wall panel drawings and Bill of materials.
- Drawings Analysis and Design of Residential and Office buildings using Light Gauge and Structural Steel.

❖	Company	Wecom Engineering Pte Ltd.,	-Singapore
	Client	: Shell Eastern Petroleum (Pte) Ltd., Singapore	
	Project	: Shell ECC-Houdini Singapore	
	Designation	: QA/QC cum Designer (Piping)	
	Duration	: May 2007 – Aug 2008	
	Technology	: Auto Cad-2007	

Accomplishments:

- Preparation of Design Documents & Detail checking of Layout and Steel detail drawings, Fabrication drawings, Wall panel drawings and Bill of materials.
- Preparation of Line List, Equipment list, P&ID, Tank farm, piping Supports, Special Supports & Parts, Nozzle Orientation Drawings act.
- Responsible for updating the quality records, material traceability, incoming materials and line inspection reports.
- Ensure the calibration status of the instruments before use in this project.
- Responsible for measuring the pipe diameter, length and thickness.
- Responsible to maintain the material traceability records. Responsible and check the 90 and 45 angle elbows.
- Preparation and maintained material traceability records (EX: length, with, thickness, fittings) based on the spool drawings.
- Responsible for updating the quality records & humidity records and maintained of process checklist, pressure gauge records and instruments calibration records used by MS Office-2003.

❖	Company	SPB Projects and Consultancy Ltd.,	- Chennai, India
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Major Projects

1.Project	: Gujarat Minerals Development Corp. Ltd, Gujarat, India
2.Project	: Waste Water treatment facilities (Unit 8400), Tamilnadu, India.
3.Project	: Khurais Central Processing(Revamping) ARAMCO.,UAE.
Duration	: Oct 2005 April 2007
Role	: Designer (Piping)
Technology	: AutoCAD, Solid works

Accomplishments:

- Preparation of offshore piping related engineering design, layout & drafting, material estimation. Preparation of Piping & Instrumentation Diagram. Preparation of isometric drawings based on AutoCAD-2006.
- Preparation of field materials and item codes according to the pipe drawings. Preparation of spools and joints diagram based on the piping lines.
- Preparation of Process Flow Diagram, Material Selection Diagram and Design Pressure and Temperature Diagram.
- Preparation and drawn all Interconnection diagrams, Spools fitting with flanges such as elbows, Data sorting and field verification of various Pump Stations, Metering Stations, Pressure Reducing Stations (PRS) of Petrochemical Plants.
- Preparation of pipe support and section details on Auto CAD. Preparing and maintaining of various IQMS documents. Equipment and Instrument cells creation.

❖ Company Aqua sub Engineering unit-II, - Coimbatore -India.

Designation : **Asst.Design Engineer. (Govt.apprentice trainee).**

Technology : **AutoCAD-2004**

Duration : **Oct 2004 to Oct 2005.**

Accomplishments:

- Preparation of Electrical motor components Diagram. Preparation of Process and line Materials Diagram used package by auto cad-2004.
- Preparation of incoming storage area diagram with materials arrangement diagram. Preparation of 2Dimension drawings based on the electric motor equipment.
- Preparing and maintaining the Detail drawings, Sectional Dwgs using variable least of dimensions for 1st angle projection and 3rd angle projection.
- Preparation of drawn the electrical equipment and line routing. Preparation the dwgs for CNC machining centre and turning centre jig, fixture preparation.
- Drawn by electrical motor components with tolerance (EX: motor body, casing, centre rods, yoke, cap, sleeve, dies and other motors components) used the technology cad-2004.
- Preparation of Rework components with correct tolerance based on the dimensions. Preparation of development components drawings and used tools diagrams.

Training Program Attended:

- ✓ Bim (MEP) Basic and Advance courses by Autodesk
- ✓ Safety Orientation Course for general trade (SOC) from NTUC Learning Hub
- ✓ Safety Orientation Course for Manhole (SOC) from NTUC Learning Hub
- ✓ Safety Induction Course (SIC) conducted by ABB for ECC Shell Project Singapore.
- ✓ Laminator Certificate from Sarplast for ECC Shell Project Singapore
- ✓ Pipe fitter Low Leavy course
- ✓ Copper pipe brazer training course

Educational & Qualifications:

May 2015	Bachelor Degree BBA, Annamalai University, Annamalai Nagar, India. Marks: 62.5 %
April 2004	Diploma Engineer Mechanical, SKAP Polytechnic, Manamadurai, Sivaganga (dist), India. Marks: 77.50%

April 2002 **Higher Secondary Certificate,**
De Britto hr.sec.school,Devakottai,
Sivaganga (dist), India.
Marks: **75.65%**

Extra Qualifications:

Having Certification for Non Destructive Testing (NDT) Level II from ASNT (American Society for Non Destructive Testing) As Per ASNT- SNT-TC-1A In,

- Liquid penetrated testing (PT)
- Magnetic particle testing (MPT)
- Ultrasonic testing (UT)
- Radiographic testing (RT)

TECHNICAL EXPERTISE:

❖ **Modeling & Drafting**

- Auto CAD 2018
- Revit (MEP) 2016, Navisworks (Support for BIM Model)
- Solid Works-2010
- Microstation V8i

❖ **Application**

- Micro soft office (MS Excel, MS Word, Power point)

Addendum:

Father name	: Rayappan,	Mother Name	: Fathima mary
Date of Birth	: 24-07-1985	Gender	: Male
Nationality	: Indian	Languages Known	: English & Tamil
Contact No	: +65-90516163	Passport number	: F4892270
Place of issue	: Tiruchirappalli	Working Days/Week	: 5.5 days
Current salary	: S\$10750	Expected salary	: S\$ 11750-12000
Residential Status	: Employment Pass	Fin No	: G8015821T

DECLARATION

I hereby solemnly affirm that the above-mentioned details are true to the best of my knowledge and belief.

Yours Truly,

(R.JESU RAJ)